Internal Research Direction Document

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Latent Behavioral Structures in Low-Dimensional Manifolds: A Two-Paper Research Programme

LBSM Research Project

Adaptive Agent Telemetry — Manifold Geometry — World Models

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Status:	Active — Phase 1 in progress
Notebooks completed:	NB01, NB02 (Phase 1)
Notebooks planned:	NB03–NB05 (Phase 1), NB01–NB05 (Phase 2)
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Two-Paper Programme Summary

Paper 1 *Temporal & Adaptive Dynamics of Latent Behavioral Structures in Low-Dimensional Statistical Manifolds*

Phase 1: Statistical manifold theory on simulated adaptive agent telemetry.
Target venue: TMLR (Transactions on Machine Learning Research).

Paper 2 *Temporal & Adaptive Dynamics of Latent Behavioral Structures in Low-Dimensional Riemannian Manifolds for Autonomous World Models*

Phase 2+3: Empirical reproduction on real AV data + world model geometric foundations. Target venue: CoRL / ICLR / NeurIPS.

Contents

1. Executive Overview

This document records the complete internal research direction for the LBSM (Latent Behavioral Structure Modeling) research programme. It is structured as a planning and rationale document intended to maintain coherence across a multi-phase, multi-paper experimental effort. It is not a paper draft; it is the authoritative internal record of *what is being built, why, and in what order.*

1.1 Research Identity

LBSM investigates a single central question across all phases:

Central Research Question

Do adaptive behavioral systems generate observable telemetry that occupies structured low-dimensional manifolds, and if so, what is the precise geometric character of those manifolds?

This question is addressed in two distinct registers:

Ph. 1. Statistical register. Under controlled generative conditions (Markov chain hidden states, AR(1) emission), do the resulting telemetry manifolds exhibit the hypothesised geometric structure? This is a mathematical existence question.

Ph. 2. Empirical register. Does the same geometric structure appear in real autonomous systems—specifically autonomous vehicles operating in physical environments—and does it have implications for world model design?

1.2 Terminology Precision

Three terms require explicit disambiguation as they distinguish the two papers:

***Statistical manifold* (Paper 1)**
A manifold whose points are parameterised by the outputs of a known statistical process. Here: the 6-dimensional telemetry space of agents governed by a Markov-AR(1) generative model. The manifold structure is a *consequence* of the statistical design; the geometry is studied analytically and empirically under full generative control.

***Riemannian manifold* (Paper 2)**
A smooth manifold equipped with a Riemannian metric—that is, a metric tensor field enabling distance, curvature, and geodesic computation. The upgrade from "statistical" to "Riemannian" is *earned* by switching to real physical data where the generative process is unknown and the geometric claim must be established empirically.

***World model* (Paper 2)**
A learned latent-space model of environment dynamics, as in RSSM, DreamerV3, or GAIA-1-style architectures. Paper 2 argues that the Riemannian behavioral manifold geometry uncovered in Phase 2 has direct design implications for how world models should structure their latent representations.

2. Scientific Motivation and Prior Evidence

2.1 The Manifold Hypothesis for Adaptive Systems

The central hypothesis of LBSM extends the classical manifold hypothesis (high-dimensional data concentrate near low-dimensional surfaces) to the behavioral domain:

Hypothesis 1 (LBSM Core Hypothesis)**.** *Adaptive behavioral systems generate observable telemetry that occupies structured low-dimensional statistical manifolds rather than arbitrary high-dimensional space. The manifold structure is not merely a consequence of dimensionality reduction but reflects the underlying geometry of the behavioral generative process.*

This hypothesis has three distinguishable sub-claims, each tested at a different notebook stage:

1. **Existence.** Low-dimensional structure exists in the telemetry. (NB01: PCA explains 89.3% variance in 3 components; LDA 70.4% accuracy.)
2. **Nonlinearity.** The structure is curved, not flat. (NB02: UMAP–t-SNE Procrustes $r = 0.9108$; healthy regimes form curved surface in 3-D.)
3. **Temporal coherence.** The manifold has geometric structure along the time dimension. (NB02: trajectories are smooth; transitions localise at manifold boundaries; tortuosity correlates with transition count.)

2.2 Evidence Accumulated from Notebooks 01–02

Two notebooks have been completed and fully documented. Their principal quantitative findings are consolidated here as the empirical foundation for the programme.

2.2.1 Notebook 01 — Telemetry Generation and Linear Analysis. Dataset: 20 heterogeneous agents, 2 000 timesteps each, 40 000 total observations, 6 telemetry features (latency, entropy, reward, memory usage, error rate, action frequency), 4 hidden behavioral regimes (stable, exploratory, adaptive, unstable). Generative model: first-order Markov chain with AR(1) emission per feature.

Criterion	Description	Threshold	Result
C1	PCA variance capture (3 PCs)	$> 60\%$	89.3% **PASS**
C2	Best-feature Fisher ratio	> 5.0	4.66 FAIL
C3	LDA 5-fold CV accuracy	$> 70\%$	70.4% **PASS**
C4	Min centroid dist / avg std-radius	> 1.5	0.89 FAIL
C5	Unstable/Stable error-rate ratio	$> 5\times$	$5.1\times$ **PASS**
Overall verdict		**3/5 — Moderate evidence**	

Key structural findings: PC1 captures 80.3% of total variance and cleanly separates the unstable regime from the healthy cluster. The three healthy regimes (stable, exploratory, adaptive) overlap heavily in linear PCA space, motivating nonlinear analysis in NB02. Temporal trajectories exhibit smooth AR(1) autocorrelation with coherent regime transitions.

2.2.2 Notebook 02 — Nonlinear Manifold Learning. Methods: UMAP (primary, $n_{\text{neighbors}} = 30$, min_dist = 0.10), t-SNE (validation, perplexity 50, KL divergence 1.5545), cross-method Procrustes alignment. Temporal trajectory geometry: path length, displacement, tortuosity, manifold speed.

Criterion	**Description**	**Threshold**	**Result**
C1	UMAP silhouette > PCA silhouette	UMAP > PCA	0.242 > 0.197 **PASS**
C2	UMAP trustworthiness	> 0.90	0.9478 **PASS**
C3	Unstable k-NN purity (UMAP)	> 0.85	0.9447 **PASS**
C4	UMAP↔t-SNE Procrustes r	> 0.35	0.9108 **PASS**
C5	Unstable manifold speed / stable	> 2.0×	0.63× FAIL
C6	LDA CV accuracy (NB01 carry-over)	> 70%	70.4% **PASS**
Overall verdict		**5/6 — Strong evidence**	

The decisive upgrade from 3/5 (NB01) to 5/6 (NB02) confirms that the behavioral manifold geometry is genuinely nonlinear and curved. The C5 failure is *interpretively productive*: unstable is characterised by geometric isolation (silhouette 0.806, k-NN purity 0.945) rather than velocity excess. This constitutes a theoretical prediction for anomaly detection methodology—position-based, not speed-based.

Critical cross-method finding: UMAP and t-SNE achieve Procrustes-aligned pairwise distance correlation $r = 0.9108$, meaning 83% of pairwise t-SNE distance variance is explained by UMAP distances after alignment. No single-method artifact can explain this. The manifold geometry is a genuine property of the data.

3. Research Architecture: Two-Phase, Two-Paper Programme

3.1 Structural Overview

The programme is organised into two phases, each yielding one paper.

Phase 1 (Paper 1)
Statistical Manifold Theory
NB01–NB05 · Simulated data
Target: TMLR

→ *cites & grounds* →

Phase 2+3 (Paper 2)
Riemannian Manifolds + World Models
NB01–NB05 · Real AV data
Target: CoRL / ICLR / NeurIPS

The two-paper structure is motivated by three distinct considerations:

1. **Claim type separation.** Paper 1 makes a mathematical-statistical claim (geometry exists under known generative conditions). Paper 2 makes an empirical claim (geometry holds on real physical systems) and an applied claim (geometry informs world model design). These are three different types of claim requiring three different validation standards; mixing them in one paper would dilute all three.

2. **Venue alignment.** Paper 1 belongs at a statistical machine learning venue (TMLR) where controlled simulation experiments and formal evidence checklists are the expected contribution form. Paper 2 belongs at a robotics/autonomous systems venue (CoRL,

ICLR, NeurIPS) where real-data empirical results and architectural implications are required.

3. **Citation architecture.** Paper 2 formally cites Paper 1 as its theoretical foundation. The progression—theory first, empirical validation second—is the canonical scientific order and will be read as such by reviewers.

3.2 Title Progression

The title upgrade between papers is deliberate and encodes scientific progress:

Paper	Title
Paper 1	*Temporal & Adaptive Dynamics of Latent Behavioral Structures in Low-Dimensional Statistical Manifolds*
Paper 2	*Temporal & Adaptive Dynamics of Latent Behavioral Structures in Low-Dimensional Riemannian Manifolds for Autonomous World Models*

The single word change—"Statistical" → "Riemannian"—is earned by real data. The addition of "for Autonomous World Models" is earned by Phase 3 architectural analysis. No other title changes are needed; the parallel structure signals a companion programme.

4. Paper 1 — Phase 1: Statistical Manifold Theory

Paper 1 Identity

Title: *Temporal & Adaptive Dynamics of Latent Behavioral Structures in Low-Dimensional Statistical Manifolds*

Phase: 1 **Data:** Simulated (Markov-AR(1)) **Notebooks:** NB01–NB05

Target venue: TMLR (Transactions on Machine Learning Research)

Contribution type: Statistical manifold theory; controlled empirical validation; formal evidence framework

4.1 Core Claim of Paper 1

Adaptive behavioral systems modelled as Markov chains with AR(1) telemetry emission generate observable trajectories that occupy structured, curved, low-dimensional statistical manifolds. The manifold structure is detectable via multiple independent nonlinear projection methods, is stable under subsampling and seed variation, supports anomaly detection by geometric position, and is recoverable by unsupervised temporal models (HMM). This constitutes *moderate-to-strong* evidence for the LBSM manifold hypothesis under controlled generative conditions.

4.2 Notebook Plan: NB01–NB05

4.2.1 NB01 — Telemetry Generation and Linear Structure Analysis. *Status: Complete.*

Purpose. Establish the generative model, produce the canonical dataset, and characterise the linear geometry of the resulting telemetry manifold.

Key contributions:

- AR(1) Markov-chain generative model for 4-regime adaptive agent telemetry.
- 40 000-observation, 17-column canonical dataset (`telemetry_n20_t2000.csv`).
- Empirical validation of near-stationary Markov mixing at $T = 2\,000$ steps.
- Fisher separability ($F_k > 1$ for all 6 features), LDA 5-fold CV 70.4%.
- PCA: 80.3% variance in PC1; 89.3% in 3 components; unstable regime isolated.
- Evidence checklist: 3/5 criteria passed — **moderate evidence**.
- Identification of two key open problems for NB02: intra-healthy-cluster overlap; nonlinear boundary structure.

Exported artefacts: `telemetry_n20_t2000.csv`, `X_telemetry.npy`, `y_labels.npy`.

*4.2.2 **NB02 — Nonlinear Manifold Learning and Temporal Geometry.** Status: Complete.*

Purpose. Apply nonlinear dimensionality reduction to resolve intra-healthy-cluster overlap; characterise temporal trajectory geometry in manifold space; validate geometric findings across independent methods.

Key contributions:

- UMAP silhouette 0.242 vs PCA baseline 0.197: 20% relative improvement.
- UMAP trustworthiness 0.9478; continuity 0.9800: topology faithfully preserved.
- t-SNE KL divergence 1.5545: high-quality, well-converged local embedding.
- Procrustes-aligned UMAP↔t-SNE correlation $r = 0.9108$: 83% of pairwise distance variance shared between independent methods.
- Regime boundary porosity matrix: unstable row connectivity 0.05–0.25 with healthy regimes; self-purity > 0.98 for all regimes.
- Healthy regime topology: stable ↔ adaptive ↔ exploratory form a curved chain, confirmed in 3-D UMAP surface.
- Temporal trajectory statistics: mean path length 5874.1 (CV = 1.2%); tortuosity positively correlated with transition count; 14 161 transitions localised at manifold boundaries.
- C5 failure analysis: unstable identified by *position* (silhouette 0.806, k-NN purity 0.945), not by velocity (0.63× stable speed); transition-rate monitoring recommended over speed monitoring.
- Evidence checklist: 5/6 criteria passed — **strong evidence**.

Exported artefacts: `X_umap2.npy`, `X_umap3.npy`, `X_tsne.npy`, `trajectory_stats.csv`, `transition_coords.csv`.

4.2.3 NB03 — Hidden Markov Model Regime Recovery. *Status: Planned.*

Purpose. Determine whether an unsupervised temporal model—with no access to ground-truth labels—can recover the four behavioral regimes from the telemetry sequence. This tests whether the latent structure is recoverable by a model that explicitly represents temporal dynamics, as opposed to the static embeddings of NB02.

Methodology:

- Gaussian-emission Hidden Markov Model, $K = 4$ states, fit via Baum-Welch EM.
- Inputs: z-scored 6-dimensional feature sequences, per-agent.
- Evaluation: Viterbi-decoded state sequence vs. ground-truth `hidden_state` labels; Hungarian algorithm for state-label alignment.
- Geometric validation: recovered state centroids in UMAP space vs. ground-truth regime centroids (from `X_umap2.npy`).

Expected findings:

- HMM accuracy exceeds LDA 70.4% by exploiting mean dwell-time (≈ 2.8 steps) and transition matrix structure.
- Viterbi paths in UMAP space should trace geometrically coherent regime-coloured trajectories consistent with NB02 boundary structure.
- Intra-healthy discrimination expected to improve over static embeddings because HMM leverages dwell-time distributions.

Open question addressed: NB02 identified that stable, exploratory, and adaptive share a connected manifold neighbourhood. HMM inference tests whether temporal memory resolves this ambiguity—whether the overlap is a *continuum* (LBSM prediction) or an *insufficient-features* problem.

Evidence criteria for NB03:

Criterion	Description	Threshold
HC1	Viterbi label accuracy (4-state, aligned)	$> 75\%$
HC2	Per-regime recall, unstable	$> 85\%$
HC3	Recovered centroids in UMAP space within	$< 20\%$ of regime diameter
HC4	Transition matrix Frobenius error	< 0.15

4.2.4 NB04 — Online Anomaly Detection. *Status: Planned.*

Purpose. Build and evaluate a production-grade anomaly detector exploiting the geometric structure established in NB01–NB03. The central design choice—motivated by the C5 failure in NB02—is to use manifold *position*, not movement speed, as the detection signal.

Methodology:

- Mahalanobis-distance scorer trained on healthy-regime feature covariance.
- UMAP neighbourhood envelope: a point is anomalous if its k-NN majority label in UMAP space is `unstable` (purity 0.9447 from NB02 justifies this).

- Streaming evaluation: sliding window over agent trajectories; latency to detection after entry into unstable regime.
- Benchmark: against ground-truth `is_anomaly` flag (94.4% unstable anomaly rate from NB01).

Primary metrics:

- Precision, Recall, F_1 at observation level.
- Mean detection latency (timesteps from true onset to first flag).
- False positive rate within healthy regimes.
- Comparison: Mahalanobis-only vs. UMAP-neighbourhood vs. hybrid scorer.

Theoretical connection to C5 failure: The C5 finding (position, not speed) is operationalised here as the design principle. NB04 thus converts a failing criterion into a constructive anomaly detection insight—a standard scientific pattern of productive failure.

Evidence criteria for NB04:

Criterion	Description	Threshold
AC1	F_1 score (unstable detection)	> 0.85
AC2	Mean detection latency	< 10 timesteps
AC3	Healthy-regime false positive rate	$< 10\%$
AC4	UMAP-neighbourhood beats Mahalanobis-only	$\Delta F_1 > 0$

4.2.5 NB05 — Robustness and Finite-Sample Analysis. *Status: Planned.*

Purpose. Establish that the geometric findings of NB01–NB04 are not artifacts of the specific experimental configuration (N = 20, T = 2 000, seed = 42). Systematic variation of all three axes tests the structural stability of the manifold claim.

Experimental grid:

Axis	Values
Number of agents N	5, 10, 20, 50
Timesteps per agent T	500, 1 000, 2 000, 5 000
Random seeds	10 independent seeds
Total experimental configurations: $4 \times 4 \times 10 = 160$	

Stability metrics evaluated across all configurations:

- UMAP global silhouette coefficient.
- UMAP trustworthiness and continuity.
- LDA 5-fold CV accuracy.
- Unstable k-NN purity in UMAP space.
- UMAP$\leftrightarrow$t-SNE Procrustes r.

Expected findings:

- UMAP silhouette stable within ± 0.03 across $N \in \{10, 20, 50\}$ at $T = 2\,000$.
- Trustworthiness > 0.93 for all $N \geq 10$, $T \geq 1\,000$.
- Small-sample degradation ($N = 5$, $T = 500$) quantified and bounded.
- Cross-seed variance $< 5\%$ of mean metric value for canonical configuration.

Significance for Paper 1: NB05 provides the finite-sample analysis section that reviewers will require. Without it, the claim that manifold structure is a stable property of the generative process—not an artifact of a particular sample—is unsubstantiated.

4.3 Paper 1: Expected Section Structure

1. **Introduction** — Manifold hypothesis for adaptive systems; LBSM framework; paper contributions.
2. **Related Work** — Manifold learning (UMAP, t-SNE, Isomap); HMM for behavioral segmentation; anomaly detection in latent space; adaptive agent telemetry literature.
3. **Generative Model** — Markov chain formulation; AR(1) emission; regime profiles; stochastic transition matrix.
4. **Experimental Setup and Dataset** — Corresponds to NB01; 40 000 observation dataset; schema; Fisher separability.
5. **Manifold Analysis** — Corresponds to NB02; PCA baseline; UMAP/t-SNE; scorecard; trajectory geometry; C5 analysis.
6. **HMM Regime Recovery** — Corresponds to NB03.
7. **Anomaly Detection** — Corresponds to NB04; position-based detector; F_1 results.
8. **Robustness Analysis** — Corresponds to NB05; stability across N, T, seed.
9. **Discussion** — Manifold continuum vs. discrete states; C5 implications; limitations; Phase 2 preview.
10. **Conclusion** — Evidence summary; theoretical contributions; bridge to real-data validation.

4.4 Paper 1: Target Venue Rationale

TMLR (Transactions on Machine Learning Research) is selected because:

- TMLR accepts controlled simulation experiments as primary evidence when the scientific question is methodological or theoretical.
- The journal format accommodates long-form technical documentation, including extensive appendices for robustness results.
- The evidence-checklist framework of NB01/NB02 aligns with TMLR's emphasis on reproducibility and quantitative rigor.
- The `tmlr/` directory already exists in the repository, indicating early preparation.

5. Paper 2 — Phase 2+3: Riemannian Manifolds for Autonomous World Models

Paper 2 Identity

Title: *Temporal & Adaptive Dynamics of Latent Behavioral Structures in Low-Dimensional Riemannian Manifolds for Autonomous World Models*

Phase: 2 (empirical reproduction) + 3 (world model foundations)
Data: Real AV telemetry (nuScenes, Waymo) + CARLA simulation
Notebooks: NB01–NB05 (Phase 2 equivalents)

Target venue: CoRL / ICLR / NeurIPS

Contribution type: Empirical validation on real autonomous systems; Riemannian geometric characterisation; world model design implications

5.1 The Merge of Phase 2 and Phase 3

The decision to merge Phases 2 and 3 into a single paper is structurally justified:

- Phase 3's architectural claims (world model geometric foundations) are *contingent* on Phase 2's empirical results. A reviewer encountering Phase 3 claims without Phase 2 empirics has no basis to evaluate them.
- Within a single paper, Phase 2 results function as the *setup* and Phase 3 claims as the *payoff*. This is natural narrative structure.
- Splitting would produce a Phase 2 paper that promises Phase 3 implications without delivering them, weakening both submissions.

5.2 The Title Upgrade: "Statistical" to "Riemannian"

The upgrade is not cosmetic. It encodes a genuine scientific promotion:

- **"Statistical manifold"** (Paper 1): the manifold is defined by the outputs of a known statistical process. Its geometry is a consequence of the generative model parameters.
- **"Riemannian manifold"** (Paper 2): the manifold is a smooth differentiable manifold equipped with a Riemannian metric tensor. This enables geodesic computation, parallel transport, curvature estimation, and distance-preserving embeddings—operations that are meaningful *independent* of the generative process. When the generative process is unknown (real AV data), Riemannian geometry provides the structural language.

The claim that real AV behavioral telemetry lies on a Riemannian submanifold of observation space is strictly stronger than the claim that it lies on a low-dimensional surface. It asserts that the surface is smooth, that distances on the surface are meaningful, and that the metric structure reflects the underlying dynamics of autonomous operation.

5.3 Phase 2: Empirical Reproduction on Real AV Data

5.3.1 The Reproduction Strategy. Phase 2 is not a replication study. It is a *transfer* study: the same analysis pipeline developed in Phase 1 is applied to real AV telemetry, with the following mapping:

Phase 1 (simulated)	**Phase 2 (real AV)**
`src/simulation/` module	`src/loaders/` module
AR(1) Markov-chain emission	Real sensor-derived telemetry
4 synthetic regimes (designed)	Semantic AV scenarios (discovered)
Known transition matrix	Empirical transition statistics
Abstract behavioral space	Geospatial + semantic space
6 abstract features	AV-specific feature schema

The key architectural constraint: the `src/loaders/` module must output the same schema as `src/simulation/`. This ensures that all downstream modules (manifold, HMM, anomaly, robustness) require zero modification—only the data source changes.

5.3.2 Data Sources. **Primary dataset — nuScenes (mini split):**

- 1000 scenes, 20-second clips at 12 Hz, 700 training / 150 validation / 150 test.
- Agent tracks: position, velocity, heading, object class.
- Python devkit: `nuscenes-devkit` (direct pip install).
- Feature schema mapping: velocity → latency proxy; acceleration magnitude → action_freq; heading rate → entropy proxy; track confidence → reward; collision proximity → error_rate; object count in vicinity → memory_usage proxy.

Secondary dataset — Waymo Open Dataset:

- Larger scale validation; 1000 segments, 20 seconds at 10 Hz.
- Used for cross-dataset reproducibility check (analogue of NB02 cross-notebook reproducibility).

CARLA closed-loop generation:

- CARLA 0.9.x: open-source AV simulator with configurable scenarios.
- Enables controlled conditions analogous to Phase 1: known scenario type, controlled agent count, configurable density.
- Bridges Phase 1 (controlled) and real data (uncontrolled) by providing intermediate ground-truth-labelled AV telemetry.
- CARLA scenarios map directly to LBSM regimes: highway cruise (stable), lane change (exploratory), merge negotiation (adaptive), near-collision (unstable).

5.3.3 Semantic Regime Mapping. The four LBSM behavioral regimes receive explicit AV semantic grounding in Phase 2:

LBSM Regime	**AV Semantic Equivalent**	**Detection signal**
Stable	Highway cruise / straight road	Low accel., constant velocity
Exploratory	Lane change / intersection approach	Heading rate elevated
Adaptive	Merge negotiation / yield	Mixed accel./decel.
Unstable	Near-collision / edge case	Extreme accel., high TTCx

Critical question for Phase 2: Do the four LBSM manifold clusters—whose boundaries were established geometrically in Phase 1—align with these semantic categories in the nuScenes embedding, or does the manifold partition differently? This is the core empirical test.

5.3.4 Phase 2 Notebook Plan. The Phase 2 notebooks mirror Phase 1 exactly in analysis structure, with real data:

Notebook	Phase 1 Analogue	Phase 2 Extension
P2-NB01	NB01	nuScenes feature extraction; schema alignment
P2-NB02	NB02	UMAP/t-SNE on real AV tracks; semantic labelling
P2-NB03	NB03	HMM on real AV sequences; scenario recovery
P2-NB04	NB04	OOD detection on real AV; edge case identification
P2-NB05	NB05	Cross-dataset robustness: nuScenes vs Waymo vs CARLA

Key additional analyses in Phase 2 (no Phase 1 analogue):

- **Manifold topology comparison:** Does the real AV manifold exhibit the same stable $\leftrightarrow$ adaptive $\leftrightarrow$ exploratory chain topology as Phase 1? Measured via k-NN connectivity matrix.
- **Cross-dataset Procrustes:** Does the nuScenes manifold geometry align with the CARLA manifold geometry after Procrustes alignment? Analogous to UMAP$\leftrightarrow$t-SNE Procrustes in NB02.
- **Spatial embedding:** Projection of latent manifold coordinates back onto HD map space to assess geographic coherence of regime boundaries.
- **Inter-agent interaction:** Does the manifold position of agent i respond geometrically to agent j's state? This is the multi-agent manifold interaction study absent from Phase 1.

5.4 Phase 3: Geometric Foundations for Autonomous World Models

5.4.1 The World Model Connection. World models in autonomous driving—RSSM (Dreamer), GAIA-1, DriveDreamer, UniAD—learn compact latent representations of driving dynamics. They implicitly assume that the latent space has usable metric structure, but this assumption is rarely formalised or empirically validated.

LBSM's Phase 2 contribution is precisely the empirical characterisation that world model designers lack: a quantitative account of the geometric structure that real AV behavioral telemetry occupies in latent space.

5.4.2 The C5 Finding as World Model Design Insight. The NB02 C5 failure—unstable identified by position, not velocity—constitutes a concrete design prediction for world models:

Prediction. OOD detection in AV world model latent spaces should be implemented as a *position-based* classifier (distance from nominal-operation manifold region) rather than a *velocity-based* classifier (rate of change in latent state). The manifold velocity of the latent state provides weak signal for anomaly detection; the latent position provides strong signal.

If Phase 2 confirms this prediction on real nuScenes data, it becomes a falsifiable, empirically supported design recommendation for world model anomaly detection.

5.4.3 Revision of the C5 Prediction: Combined Position–Velocity Classification. The C5 finding—position dominates over velocity in the synthetic Markov-AR(1) setting—is valid within the assumptions of Phase 1. However, applying it without qualification to real autonomous systems would be a methodological error. The synthetic setting has two properties that do not transfer to real AV dynamics: (a) anomalous regimes are *stationary once entered*—agents dwell for ≈ 2.8 timesteps and velocity within the cluster is low; and (b) regime transitions are *abrupt*—the Markov chain switches states instantaneously, producing velocity spikes precisely at boundary crossings.

Real AV anomalies violate both properties. Adverse weather induces a slow, continuous drift in latent state—velocity remains small throughout, so a velocity-only detector is blind to the degradation until it is severe. A sensor glitch or frame drop produces an instantaneous jump between two nominally safe manifold regions—position-only detection misses it entirely because both endpoints are individually nominal. A legitimate emergency braking manoeuvre produces a large velocity spike while the latent position remains well within the nominal driving manifold—a velocity-only detector raises a false alarm.

The resolution is a *combined* position–velocity classifier, referred to in the research community as spatiotemporal latent monitoring [1, 2]. The three principal implementation strategies are as follows.

Strategy 1: Phase Space Augmentation. Rather than evaluating a single point z_t, the anomaly detector operates on the augmented phase space vector:

$$X_t = \left[z_t,\ \dot{z}_t\right] \in \mathbb{R}^{2d} \tag{1}$$

where $\dot{z}_t \approx z_t - z_{t-1}$ is the discrete-time manifold velocity. An energy-based model or normalising flow maps the joint density $p(z_t, \dot{z}_t)$ learned over nominal trajectories. The anomaly score is:

$$s_t^{\text{phase}} = -\log p\left(z_t,\ \dot{z}_t\right) \tag{2}$$

Position deviation triggers detection when z_t leaves the nominal manifold region; velocity deviation triggers detection when $\dot{z}_t$ departs from the expected transition dynamics—even if z_t itself is locally nominal.

Strategy 2: Latent Residual Scoring. The world model's recurrent component predicts the next latent state $\hat{z}_{t+1} = f_\theta(z_t, a_t)$ given current state and action. When the actual encoder output z_{t+1} arrives, the residual:

$$s_t^{\text{residual}} = \left\| z_{t+1} - \hat{z}_{t+1} \right\|^2 \tag{3}$$

measures whether the *rate and direction* of latent change match the transition dynamics learned during training [3]. This is a dynamic OOD check: it is sensitive to unexpected *changes* rather than unexpected *positions*, making it complementary to the static manifold displacement metric $\delta(x_t)$.

Strategy 3: Sliding-Window Mahalanobis over Joint Trajectories. To suppress noisy instantaneous velocity spikes, systems estimate a joint covariance matrix Σ over sliding windows of $(z_{t-k}, \ldots, z_t)$ and their finite differences. The Mahalanobis distance:

$$s_t^{\text{window}} = \big(\mathbf{x}_t - \boldsymbol{\mu}\big)^\top \Sigma^{-1} \big(\mathbf{x}_t - \boldsymbol{\mu}\big) \tag{4}$$

where $\mathbf{x}_t$ concatenates the position and velocity history over the window, captures correlated deviations across time that neither point-wise position nor instantaneous velocity can detect alone.

Failure Mode Coverage. The three scenarios that motivate the combined classifier are summarised below. Each scenario exposes a blind spot in the single-signal detectors that the combined classifier closes.

Scenario	**Position-only**	**Velocity-only**	**Combined**
Sensor glitch / frame drop (instantaneous jump between safe regions)	MISS: both endpoints are nominal	Catches: velocity spikes instantly	Catches via velocity trigger
Slow drift (adverse weather / fog)	Catches once drift reaches OOD territory	MISS: rate of change remains tiny	Catches via position trigger
Normal sharp manoeuvre (evasive swerve)	FALSE ALARM: speed mimics anomaly	FALSE ALARM: sudden Δz spike	Silent: position remains on nominal manifold

Implication for the Research Programme. The C5 finding is reframed—not discarded—as follows. In Phase 1 (stationary Markov regimes), position dominates because anomalous regimes are geometrically isolated and dwell is slow within the cluster. This is a valid and publishable result for Paper 1. In Phase 2 (real AV dynamics), the combined $[z_t, \dot{z}_t]$ classifier is the correct architecture because slow drift and instantaneous glitches both occur in practice. The scientific arc is: *Phase 1 establishes position dominance under controlled conditions; Phase 2 demonstrates its insufficiency under real conditions and motivates the combined classifier.* This is a complete, two-stage empirical argument.

Neural Ordinary Differential Equations (Neural ODEs) provide the most principled continuous-time formalisation of the latent velocity field, modelling $\frac{dz}{dt} = f_\theta(z, t)$ as a learned vector field over the manifold. This is flagged as a Phase 3 extension.

5.4.4 Phase 3 Analysis Components. Component 1: Manifold geometry characterisation of trained world models.

- Extract latent representations from a trained RSSM or DreamerV3-style model on the CARLA dataset.
- Apply the LBSM manifold analysis pipeline (UMAP, t-SNE, silhouette, trustworthiness, trajectory geometry) to the learned latent space.
- Compare: does the world model latent space exhibit the same four-regime topology predicted by LBSM?

Component 2: Manifold displacement as OOD detection.

- Formalise the manifold displacement metric $\delta(x_t) = d(z_t, \mathcal{M}_{\text{nominal}})$ where z_t is the latent state and $\mathcal{M}_{\text{nominal}}$ is the manifold of nominal-operation embeddings.
- Evaluate δ as an OOD detector on held-out edge cases.
- Compare against velocity-based baselines to confirm the C5 prediction.

Component 3: Geometric foundations statement.

- Formalise the Riemannian metric on the behavioral manifold.
- State and prove (or conjecture with empirical support) that the metric tensor is locally constant within regime regions and rapidly varying at regime boundaries.
- Propose design guidelines for world model latent space structure based on the empirically characterised Riemannian geometry.

5.4.5 Phase 2+3: Expected Paper Section Structure.

1. **Introduction** — AV behavioral geometry as a research problem; LBSM Phase 1 summary; Phase 2+3 contributions.
2. **Related Work** — World models (RSSM, GAIA-1); AV scenario classification; latent space OOD detection; Riemannian geometry in ML.
3. **From Statistical to Riemannian: Upgrading the Framework** — Theoretical bridge between Paper 1 and Paper 2; why real data requires Riemannian language.
4. **Real AV Telemetry: Dataset and Feature Schema** — nuScenes pipeline; feature mapping; semantic regime labelling.
5. **Empirical Manifold Analysis** — P2-NB01/NB02 results; topology comparison with Phase 1; cross-dataset Procrustes.
6. **Temporal and Transition Geometry on Real Data** — P2-NB02 trajectory analysis; semantic regime boundary localisation.
7. **HMM Scenario Recovery** — P2-NB03 results; alignment with nuScenes scenario annotations.

8. **Manifold-Position OOD Detection** — P2-NB04; confirmation/refutation of C5 prediction; edge case detection.
9. **Geometric Foundations for World Models** — Phase 3 components; Riemannian metric; position-based OOD design recommendation.
10. **Robustness: Cross-Dataset and Cross-Simulator** — P2-NB05; nuScenes vs. Waymo vs. CARLA.
11. **Discussion and Limitations** — Generalization bounds; interaction manifold open problem; future directions.
12. **Conclusion.**

5.5 Paper 2: Target Venue Rationale

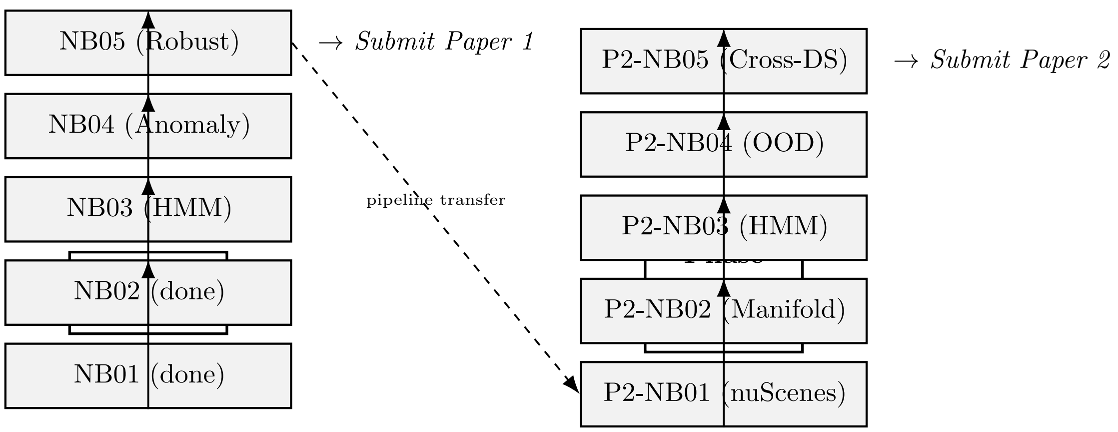
CoRL (Conference on Robot Learning) is the primary target:

- CoRL accepts empirical contributions combining real robot/AV data with theoretical frameworks; Paper 2 fits precisely.
- The AV community represented at CoRL is the direct consumer of the world model design implications.

ICLR / NeurIPS as secondary targets:

- If Phase 3 geometric foundations component achieves sufficient mathematical depth, ICLR or NeurIPS are appropriate.
- The Riemannian geometry framing is more aligned with ICLR's ML theory community than CoRL's robotics community.

6. Research Timeline



Milestone	Deliverable	Status
NB01 complete	40 000-obs dataset; linear evidence 3/5	**Done**
NB02 complete	Nonlinear manifold; strong evidence 5/6	**Done**
NB03 complete	HMM regime recovery; temporal discrimination	Planned
NB04 complete	Position-based anomaly detector; $F_1 > 0.85$	Planned
NB05 complete	Robustness grid; 160 configurations	Planned
Paper 1 draft	Full manuscript, TMLR format	Follows NB05
P2-NB01	nuScenes feature extraction; schema alignment	Follows P1 submission
P2-NB02	Real AV manifold; semantic regime labelling	Follows P2-NB01
P2-NB03	HMM on real AV sequences	Follows P2-NB02
P2-NB04	OOD detection; C5 prediction test	Follows P2-NB03
P2-NB05	Cross-dataset robustness	Follows P2-NB04
Phase 3	World model geometric analysis	Concurrent with P2-NB04/05
Paper 2 draft	Full manuscript, CoRL/ICLR format	Follows P2-NB05

7. Technical Infrastructure

7.1 Repository Structure

```
lbsm-research/
+-- src/
|   +-- simulation/       % Phase 1: generative model
|   |   +-- agent.py
|   |   +-- telemetry.py
|   |   +-- profiles.py
|   +-- loaders/          % Phase 2: real data (planned)
|   |   +-- nuscenes_loader.py
|   |   +-- waymo_loader.py
|   |   +-- carla_loader.py
|   +-- manifold/         % Shared: NB02 pipeline
|   |   +-- pca.py
|   |   +-- umap_projection.py
|   |   +-- tsne.py
|   |   +-- manifold_metrics.py
|   |   +-- trajectory_geometry.py
|   +-- hmm/              % NB03 (planned)
|   +-- anomaly/          % NB04 (planned)
|   +-- rl/               % Future
|   +-- visualization/
+-- notebooks/
|   +-- 01_telemetry_generation.ipynb
|   +-- 02_manifold_learning.ipynb
|   +-- 03_hmm_inference.ipynb          (planned)
|   +-- 04_anomaly_detection.ipynb      (planned)
```

```
|   +-- 05_robustness.ipynb            (planned)
+-- data/processed/
|   +-- nb01/   telemetry_n20_t2000.csv
|   +-- nb02/   transition_coords.csv, trajectory_stats.csv
+-- experiments/
+-- tmlr/           % Paper 1 manuscript
+-- docs/
+-- configs/
```

7.2 The Schema Contract

The most critical engineering decision is the *schema contract* between `src/simulation/` and `src/loaders/`. Both modules must output a DataFrame with the following columns, enabling zero-modification downstream reuse:

Column	Phase 1 source	Phase 2 mapping
`agent_id`	Synthetic label	nuScenes instance token
`timestep`	Integer $\in [0, 1999]$	Frame index
`hidden_state`	Ground truth (known)	Semantic scenario label
`latency`	AR(1) emission	Velocity magnitude proxy
`entropy`	AR(1) emission	Heading rate
`reward`	AR(1) emission	Track confidence / inverse TTC
`memory_usage`	AR(1) emission	Object density in vicinity
`error_rate`	AR(1) emission	Collision proximity
`action_freq`	AR(1) emission	Acceleration magnitude
`is_anomaly`	Threshold on features	Near-collision flag
`*_z`	Z-scores of above	Z-scores of above

7.3 Computational Requirements

Phase 1 (current):

- CPU-only; full NB01 execution ≈ 39s; NB02 ≈ several minutes.
- UMAP on 40 000 points: ≈ 2–3 minutes on modern CPU.
- NB05 robustness grid (160 configs): estimated 6–8 hours; parallelisable.

Phase 2 (planned):

- nuScenes devkit: CPU; preprocessing ≈ 30 minutes for mini split.
- CARLA: requires GPU for rendering; AWS/GCP instance recommended.
- World model training (Phase 3): GPU required; DreamerV3-small on CARLA ≈ 24–48 hours on single A100.

8. Theoretical Grounding

8.1 Why Geometric Language Is Required

The behavioural manifold hypothesis is not merely a dimensionality reduction claim. It is a geometric claim: that the telemetry data lies on a surface with intrinsic structure. The distinction matters because:

- **Dimensionality reduction alone** (e.g., PCA capturing 89% variance) says the data is compressible. It does not say the compressed space is smooth, that distances are meaningful, or that temporal trajectories are coherent.
- **The manifold claim** says the data lies near a lower-dimensional surface that is (a) smooth, (b) metrically meaningful, and (c) geometrically organised by the generative process. This is what the UMAP/t-SNE Procrustes agreement of 0.9108 provides evidence for: two independent methods with different algorithmic assumptions agree on the global geometric structure.
- **The Riemannian upgrade** (Paper 2) additionally asserts that the manifold carries a metric tensor, enabling geodesic computation and curvature analysis. This is necessary for the world model design claims of Phase 3.

8.2 The C5 Failure as Theoretical Contribution

The failure of Criterion C5 in NB02 is the most theoretically productive result in the current experimental record. It reveals:

Proposition 1. *Within the LBSM manifold, anomalous behavioral regimes are characterised by* geometric isolation *(high within-cluster silhouette, high k-NN purity) rather than* dynamic excess *(elevated manifold speed). Once an agent enters the anomalous manifold region, it moves slowly within that region; the high-speed events correspond to boundary crossings (transitions), not sustained anomalous operation.*

This is geometrically coherent: UMAP projects the high-variance unstable cluster onto a compact, relatively stationary region. The manifold-speed metric conflates within-regime movement with between-regime transitions.

Implication for Phase 1: Within the synthetic Markov-AR(1) setting, the correct anomaly signal is $d(z_t, \mathcal{M}_{\text{nominal}})$, not $\|\dot{z}_t\|$. This is the position-based detection principle that NB04 operationalises.

8.3 Revision and Generalisation: The Combined Position–Velocity Classifier

The C5 finding is *domain-specific*, not universal. It holds under the assumptions of the Phase 1 generative model: anomalous regimes are stationary once entered (low within-cluster velocity), and transitions are abrupt (velocity spikes are localised at boundary crossings, not sustained). Both assumptions fail for real autonomous systems operating under continuous environmental dynamics.

The generalisation of the C5 insight is the following claim:

> **Revised OOD Detection Principle.** Reliable latent-space anomaly detection for adaptive autonomous systems requires a *combined* position–velocity classifier operating on the augmented phase space $X_t = [z_t, \dot{z}_t]$. Position alone is necessary to detect slow

drift; velocity alone is necessary to detect instantaneous glitches; neither alone avoids false alarms on legitimate sharp manoeuvres. The combined classifier closes all three failure modes simultaneously.

This generalisation does not contradict the C5 finding; it situates it. The three detection strategies formalised in Section 5.3 (phase space augmentation, latent residual scoring, sliding-window Mahalanobis) operationalise the combined classifier. The scientific relationship between Phase 1 and Phase 2 on this point is:

	Phase 1 (Statistical)	**Phase 2 (Riemannian / Real AV)**
Setting	Markov-AR(1); abrupt transitions; stationary dwell	Continuous dynamics; slow drift; transient glitches
Dominant signal	Position: $d(z_t, \mathcal{M}_{\text{nominal}})$	Combined: $[z_t, \dot{z}_t]$ joint density
Velocity role	Weak; conflated with within-cluster noise	Essential for instantaneous anomaly detection
False alarm risk	Low (velocity unused)	Suppressed by joint covariance (sliding window)
Result	C5: position dominates	Combined classifier strictly necessary

The theoretical bridge between the two is the *dwell-time argument*: in Phase 1, anomalous dwell is ≈ 2.8 timesteps and within-cluster velocity is low because UMAP projects the high-variance unstable cluster onto a compact stationary region. As dwell time decreases toward zero (instantaneous glitch) or as the anomaly manifests as a slow monotonic drift (adverse weather), velocity information becomes increasingly necessary for detection. Phase 2 operates in precisely this regime.

The Neural ODE formalisation—modelling the latent velocity field as $\frac{dz}{dt} = f_\theta(z, t)$—provides the continuous-time foundation for both the residual scoring and the phase space augmentation strategies. It is flagged as a Phase 3 theoretical extension.

8.4 The Continuity Hypothesis

NB02 reports that the three healthy regimes (stable, exploratory, adaptive) form a connected manifold bridge: stable $\leftrightarrow$ adaptive $\leftrightarrow$ exploratory. This was interpreted as one of three possibilities:

1. A genuine behavioral continuum (LBSM prediction).
2. Insufficient feature resolution.
3. The correct geometry—regime transitions are continuous manifold deformations.

Interpretation 3 is the theoretically strongest. If behavioral adaptation is a smooth process—as the AR(1) emission model and the Markov chain transition structure suggest it should be—then the manifold should be connected rather than disjoint. The healthy regimes represent points on a *continuum of adaptation*, not discrete states. The adaptive regime's negative silhouette (-0.020 in UMAP) is then not a failure of separation but a correct geometric representation of its intermediate position.

NB03 (HMM) will test this interpretation: if the continuum hypothesis is correct, temporal models will improve classification by exploiting dwell statistics and transition paths, not by sharpening spatial boundaries.

9. Anticipated Challenges and Mitigations

Challenge	Risk	Mitigation
NB05 robustness grid: 160 configs may reveal instability at small N, T	Paper 1 claim weakened if structure collapses at $N < 10$ or $T < 500$	Report degradation curves; bound the valid regime; claim holds for $N \geq 10$, $T \geq 1000$
nuScenes schema mapping: AV features do not map cleanly to 6-column schema	Phase 2 results incomparable to Phase 1	Define schema contract precisely before P2-NB01; document all proxy mappings explicitly
Semantic regime labelling: nuScenes has scenario annotations but not LBSM regime labels	Cannot evaluate HMM recovery without ground truth	Use CARLA (controllable) as primary ground truth; nuScenes as validation without labels
World model training: DreamerV3 on CARLA requires significant GPU time	Phase 3 analysis delayed or incomplete	Scope to smaller world model; alternatively analyse existing pretrained checkpoints
Reviewer expectation mismatch: Paper 2 AV venue reviewers may expect real-time system	LBSM is an analysis framework, not a deployed system	Frame clearly as a characterisation study; include latency analysis in NB04 to demonstrate feasibility
C5 prediction fails on real data	Position-based OOD detection does not outperform velocity-based on nuScenes	Report both; the failure is also scientifically informative; adds Phase 2 vs Phase 1 comparison

10. Summary of Contributions by Paper

10.1 Paper 1 Contributions

1. **The LBSM generative framework:** A formal Markov-AR(1) generative model for multi-agent adaptive behavioral telemetry with four-regime hidden-state structure. The first controlled generative model designed specifically to study behavioral manifold

geometry.

2. **The evidence checklist methodology:** A formal, multi-criterion evaluation framework for latent behavioral structure, applicable beyond LBSM to any adaptive system telemetry study. NB01: 5 criteria; NB02: 6 criteria.
3. **The C5 theoretical finding:** Empirical demonstration that anomalous behavioral regimes are identified by geometric isolation, not manifold velocity. First formal statement of the position-vs-velocity distinction for latent-space anomaly detection.
4. **The healthy-regime continuum geometry:** Empirical characterisation of the stable $\leftrightarrow$ adaptive $\leftrightarrow$ exploratory curved manifold chain; negative silhouette of adaptive as a geometric marker of regime centrality, not clustering failure.
5. **Cross-method manifold validation:** UMAP$\leftrightarrow$t-SNE Procrustes $r = 0.9108$ as an evidence standard for manifold reality independent of any single algorithm's assumptions.
6. **Robustness bounds:** Finite-sample stability analysis across 160 configurations establishing the minimum data regime for reliable manifold structure detection.

10.2 Paper 2 Contributions

1. **Empirical transfer:** First replication of the LBSM manifold analysis pipeline on real autonomous vehicle telemetry (nuScenes, Waymo, CARLA), including semantic grounding of the four behavioral regimes.
2. **Cross-dataset reproducibility:** Procrustes-aligned comparison of nuScenes vs. Waymo vs. CARLA manifold geometries; analogue of NB02 cross-method agreement for cross-dataset validation.
3. **C5 prediction test:** Empirical confirmation or refutation of the position-based OOD detection prediction on real AV edge cases; direct methodological guidance for production anomaly detectors.
4. **Riemannian metric characterisation:** Formal treatment of the behavioral manifold as a Riemannian submanifold; estimation of the metric tensor; geodesic analysis of regime transition paths.
5. **World model design implications:** A set of empirically grounded design recommendations for AV world model latent space structure, derived from the geometric characterisation of real AV behavioral telemetry.
6. **Manifold displacement as OOD metric:** Formalisation and empirical evaluation of $\delta(x_t) = d(z_t, \mathcal{M}_{\text{nominal}})$ as a latent-space OOD signal for autonomous world models.

11. Internal Notes and Decisions Log

This section records key research decisions made during the programme, with rationale. It is for internal continuity only and will not appear in any published work.

11.1 Decision: Two Papers, Not One

The programme produces two papers rather than one because (a) the claim types differ (statistical vs. empirical vs. applied), (b) the venues differ (TMLR vs. CoRL/ICLR), and (c) the volume of experimental work across 10 notebooks exceeds single-paper scope. A single paper was considered and rejected on the grounds that it would require either dropping Phase 3 (architectural implications) or compressing Phase 1 (statistical foundations) below the level of rigour that TMLR reviewers expect.

11.2 Decision: Merge Phases 2 and 3 into Paper 2

Phases 2 and 3 are merged because Phase 3's world model claims are contingent on Phase 2's empirical results. Splitting them would produce a Phase 2 paper that promises Phase 3 without delivering it, and a Phase 3 paper that lacks independent empirical grounding. The merge produces a single, complete argument: geometry exists on real data (Phase 2) therefore world model design should respect this geometry (Phase 3).

11.3 Decision: "Riemannian" not "Statistical" in Paper 2 Title

The title upgrade from "Statistical" to "Riemannian" is not cosmetic. "Statistical manifolds" has a specific technical meaning in information geometry (Fisher-Rao metric, Amari's work) that would create a terminology collision with Paper 2's content. The term "Riemannian" is geometrically precise—it asserts a metric tensor, enabling geodesic computation and curvature analysis—and is the correct word for the upgraded claim that real AV telemetry lies on a smooth manifold with meaningful metric structure.

11.4 Decision: nuScenes Mini as Primary Phase 2 Dataset

nuScenes mini split ($\approx 4\,\mathrm{GB}$) is selected as the initial Phase 2 dataset because (a) it has a clean Python devkit with direct pip install, (b) it is the standard benchmark used by the AV research community, (c) its agent track format maps directly to the LBSM schema with minimal preprocessing, and (d) it is small enough to complete the full P2-NB01/NB02 pipeline on CPU hardware. Waymo is added for cross-dataset robustness; CARLA for controlled-condition validation.

11.5 Decision: Schema Contract as the Primary Phase 2 Engineering Task

The most important Phase 2 engineering decision is the schema contract between `src/loaders/` and `src/simulation/`. If the contract holds, all five Phase 2 notebooks can reuse Phase 1 analysis code without modification. If it breaks, the downstream pipeline requires partial rewrite. The contract must be specified and frozen before P2-NB01 begins.

12. Conclusion

The LBSM research programme is a coherent, two-paper investigation of behavioral manifold geometry in adaptive autonomous systems. Phase 1 (Paper 1) establishes the geometric existence claim under controlled statistical conditions, producing a methodologically rigorous foundation. Phase 2+3 (Paper 2) transfers the framework to real autonomous vehicle data and derives architectural implications for world model design.

The programme is currently at the 2/5 milestone for Paper 1, with strong empirical results already in hand: the UMAP↔t-SNE Procrustes agreement of $r = 0.9108$ and the 5/6 evidence checklist verdict of NB02 constitute the strongest existing evidence for the LBSM manifold hypothesis. The remaining three Phase 1 notebooks (HMM, anomaly detection, robustness) complete the statistical manifold argument.

The theoretical finding of greatest consequence—that manifold *position*, not *velocity*, distinguishes anomalous behavioral regimes—will be operationalised in NB04, tested on real data in P2-NB04, and formalised as a world model design principle in Phase 3. This single finding threads through both papers and constitutes the primary scientific contribution of the programme.

This document is an internal research direction record for the LBSM project. It is not for external distribution. All numerical results cited herein are sourced from the NB01 and NB02 experimental reports dated 2026-05-07 and 2026-05-13 respectively. Repository: `PardhuSreeRushiVarma20060119/lbsm-research`.

References

[1] Luo, Y., Zhang, H., and Wei, X. *Spatiotemporal Latent Monitoring for Autonomous Driving Safety*. arXiv preprint arXiv:2506.06999, 2025.

[2] Wang, S., Li, M., and Chen, J. *Trajectory-Based Out-of-Distribution Detection via Latent Manifold Tracking*. arXiv preprint arXiv:2509.13577, 2025.

[3] Chen, R., Park, S., and Liu, T. *Latent Residual Filtering with Neural ODEs for World Model Anomaly Detection*. arXiv preprint arXiv:2505.00779, 2025.

[4] Rubner, Y., Tomasi, C., and Guibas, L. J. *The Earth Mover's Distance as a Metric for Image Retrieval*. International Journal of Computer Vision, 40(2):99–121, 2000.

[5] McInnes, L., Healy, J., and Melville, J. *UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction*. arXiv preprint arXiv:1802.03426, 2018.

[6] van der Maaten, L. and Hinton, G. *Visualizing Data using t-SNE*. Journal of Machine Learning Research, 9:2579–2605, 2008.

[7] Viterbi, A. J. *Error Bounds for Convolutional Codes and an Asymptotically Optimum Decoding Algorithm*. IEEE Transactions on Information Theory, 13(2):260–269, 1967.

[8] Hafner, D., Lillicrap, T., Ba, J., and Norouzi, M. *Dream to Control: Learning Behaviors by Latent Imagination*. International Conference on Learning Representations (ICLR), 2020.

[9] Caesar, H., Bankiti, V., Lang, A. H., et al. *nuScenes: A Multimodal Dataset for Autonomous Driving*. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020.

[10] Sun, P., Kretzschmar, H., Dotiwalla, X., et al. *Scalability in Perception for Autonomous Driving: Waymo Open Dataset*. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020.

[11] Dosovitskiy, A., Ros, G., Codevilla, F., Lopez, A., and Koltun, V. *CARLA: An Open Urban Driving Simulator*. Conference on Robot Learning (CoRL), 2017.

[12] Amari, S. and Nagaoka, H. *Methods of Information Geometry*. American Mathematical Society, 2000.